

* (2 is continued) Therefore there are 98 we choose from (not 1 not the place of the first number. However we have 99 integers left to pick from. By the PP, two integers must have the same value.

* (3) ~~transform~~ transform all num to $2^a \cdot b$, remove 1 from being an option, therefore: does not consider $2^a \cdot 1, a \geq 0$!!!

Therefore the smallest number must be 100. $\square$

(4) Let $X = \{1, 2, \dots, 2n\}$

$$\text{Let } Y = \left\{ \left\lceil \frac{a}{2} \right\rceil \mid a \in X \right\} = \{1, 1, 2, 2, \dots, n, n\}$$

$$= \{1, 2, \dots, n\}$$

for any elements

~~$a_1, a_2 \in X, a_2 \in X$ that differ by 1, we can see that~~

any two numbers $a_1 \in X, a_2 \in X$ that map to the same element $b_1 \in Y$, must differ by 1:

$$\left\lceil \frac{a_1}{2} \right\rceil = \left\lceil \frac{a_2}{2} \right\rceil = b_1$$

$$\left\lceil \frac{a_1}{2} \right\rceil = \left\lceil \frac{a_1+1}{2} \right\rceil, a_2 = a_1+1, a_1/2$$

by the PP,

Since $n+1$ numbers get chosen, and $|Y| = n$, there ~~must~~ must be two integers in X that map to the same one in Y , so they must differ by 1. $\square$

(5) scratch work

$$1, 4, 7, \dots$$

$$1+3 \quad 1+3 \cdot 2$$

$$1+3 \cdot (n-1) \rightarrow n^{\text{th}} \text{ number}$$

$$1+3 \cdot ((n+1)-1)$$

$$1+3 \cdot (n) = 3n+1$$

Consider $X = \{1, 2, \dots, n\}$

Let $Y = \{\lfloor \frac{a}{2} \rfloor \mid a \in X\} = \{1, 2, \dots, n\}$

any two numbers that differ by at most 2 will map to the same number in Y . Therefore since $|Y| = n$ and $n+1$ are chosen, by the PP, two numbers must map to the same # in Y meaning they differ by at most 2. $\square$

⑥ ~~about generally~~

for any n differing by k : choose $n+1$ from $\{1, 3, \dots, k \cdot n\}$

then $Y = \{\lfloor \frac{a}{k} \rfloor \mid a \in X\} = \{1, 2, \dots, n\}$ $\square$

⑦ scratch work

$$a_1 + a_2 = m \cdot 100 + r$$

if two have same remainder, $a_1 = m_1 \cdot 100 + r_1, a_2 = m_2 \cdot 100 + r_2$

difference $\rightarrow$ is possible

remainders. if all are unique, $(m_1 \cdot 100 + r_1) + (m_2 \cdot 100 + r_2) = m \cdot 100$

then two must sum to 100

$$(m_1 + m_2) \cdot 100 + r_1 + r_2 = m \cdot 100$$

$$r_1, r_2, r_3, \dots, 100 - r_1, 100 - r_2, \dots$$

$$0 \leq r_i \leq 99 \rightarrow 100 \geq 100 - r_i \geq 1$$

100 possible unique numbers similarly $\rightarrow r_1 - r_2 = 0 \rightarrow r_1 = r_2$

Proof Define the 52 integers $X = \{x_1, x_2, \dots, x_{52}\}$

all numbers are unique, otherwise their difference is 0, which is possible by 100.

lets rewrite all numbers to be of the form:

$$x_i = m_i \cdot 100 + r_i \rightarrow m_i \geq 0, 0 \leq r_i \leq 99$$

consider the sequence $R = \{r_1, r_2, r_3, \dots, r_{52}\}$

if the numbers aren't unique, the difference of two will be divisible.

$$\begin{aligned} r_j - r_i &= 0 \rightarrow r_j = r_i & x_j > x_i & \rightarrow x_j - x_i = (m_j \cdot 100 + r_j) - (m_i \cdot 100 + r_i) \\ &\rightarrow x_i = m_i \cdot 100 + r_i & & = (m_j - m_i) \cdot 100 + (r_j - r_i) \\ &x_j = m_j \cdot 100 + r_j & & = (m_j - m_i) \cdot 100 \end{aligned}$$

In the case that all r_i are unique, the sum of some $x_i + x_j$, $i \neq j$, will be divisible by 100.

The bounds for all r_i are $\rightarrow 0 \leq r_i \leq 99$

consider a new sequence

$$100 - r_1, 100 - r_2, \dots, 100 - r_{52}$$

The bounds for any # of this new sequence is:

$$100 \geq 100 - r_i \geq 1$$

Now consider the sequence of length 104:

$$r_1, r_2, \dots, r_{52}, 100 - r_1, 100 - r_2, \dots, 100 - r_{52}$$

Since there are at most 101 different possible values for any # in the sequence (0...100), By the PP, there must be two #s that are equivalent. Since all r_i are unique, the ~~equivalent~~ equivalent #s must both come from different original sequences:

$$r_i = 100 - r_j, i \neq j$$

$$\rightarrow r_i + r_j = 100$$

$$x_i + x_j = (m_i \cdot 100 + r_i) + (m_j \cdot 100 + r_j)$$

$$= (m_i + m_j)100 + (r_i + r_j)$$

$$= (m_i + m_j)100 + 100$$

$$= (m_i + m_j + 1)100$$

□

Homework 2

Note

I misread and ended up doing some do it for understanding problems instead of the turn in ones, I figured I could include them here since it took me a long time to do them :)

12

Show by example that the conclusion of the Chinese remainder theorem (Application 6) need not hold when m and n are not relatively prime.

Consider $m = 2, n = 6$

$$\begin{aligned}x &= mp + a \\x &= nq + b\end{aligned}$$

consider the sequence:

$$\begin{aligned}a &= 1 \rightarrow \\1, 1 + 2, 1 + 4, \dots \\ \{1, 3, 5, 7, 9\} \mod 6 &= \boxed{1, 3, 5}, \boxed{1, 3, 5} \dots\end{aligned}$$

Prove that, for any $n + 1$ integers a_1, a_2, a_{n+1} , there exist two of the integers a_i and a_j with $i \neq j$ such that $a_i - a_j$ is divisible by n .

Consider rewriting each a_i in the following form:

$$a_i = np_i + r_i \rightarrow 0 \leq r_i \leq n - 1$$

There are at most n distinct possible values for r_i , but $n + 1$ elements, therefore by the PP, at least two numbers must share the same r_i :

$$\begin{aligned}r_i &= r_j, i \neq j \\a_i &= np_i + r_i, a_j = np_j + r_j \\a_i - a_j &= (np_i + r_i) - (np_j + r_j) \\&= (p_i - p_j)n + (r_i - r_j) \\&= (p_i - p_j)n\end{aligned}$$

□

16

Prove that in a group of $n > 1$ people there are two who have the same number of acquaintances in the group. (It is assumed that no one is acquainted with oneself).

Let x_i be the number of acquaintances the i 'th person has.

The most x_i can be is $x_i \leq n - 1$, in the case that one person knows everyone. Similarly, the smallest x_i can be is $0 \leq x_i$, in the case that one person doesn't know anyone.

However, if any $x_i = n - 1$, it implies that person know everything, in which case $x_j = 0$ can never occur since

everyone must know i . Conversely, if any $x_i = 0$, then no $x_j = n - 1$ since one person cannot know anyone. Therefore, the # of distinct x_i is at most $n - 1$.

Thus, there are $n - 1$ possible distinct x_i , but n people that each have an x_i associated with them. Therefore by the PP, $x_i = x_j, i \neq j$. $\square$

20

Prove that $R(3, 3, 3) \leq 17$.

Consider a graph made up of 17 people, where all the connections between people are colored either red, green, or blue. Assume this graph has a coloring such that there exists no group of 3 people who all share the same connection color with each other (so no group of 3 people who all have red/blue/green connections with each other).

First, fix a person A . They have 16 connections to everyone else. Each connection must be of a color red, green, or blue.

Consider grouping A 's connections based on their assigned color (people whose connection to A is red, people whose connection with A is green, etc.). By symmetry, the argument presented below works for any combination of colors and chosen people.

Let the # of people whose connection with A is red be denoted by r . Similarly, let the # of people whose connection with A is green be denoted by g , and finally the same with the blue group being denoted with b . It can be seen that for the graph to have the assumed coloring, $r + g + b = 16$.

If any two people B and C in this group have a red connection then a triangle is completed with A (A, B, C would all form a red triangle). Therefore, all connections in this group must either be green or blue. Since $R(3, 3) = 6$, the smallest number of people in the red group, such that no red, blue, or green triangles forms, has to be 5. In other words, $r \leq 5$.

Now consider the green group. Using the same argument used for the red group, no people in this group can be colored green, so the smallest number of people that can be in this group such that no triangles are made is 5, so $g \leq 5$.

Now consider the blue group. Exactly as before, the same argument means that there must be at most 5 people whose connection with A is blue, so $b \leq 5$.

Thus, we can see that $r \leq 5, g \leq 5, b \leq 5$, so $r + b + g \leq 15$ follows. However, for the coloring to be valid $r + g + b = 16$, which as shown is impossible. Therefore by contradiction, $K_{17} \rightarrow K_3, K_3, K_3$, implying $R(3, 3, 3) \geq 17$. $\square$

22

Prove that

$$R(\underbrace{3, 3, \dots, 3}_{k+1}) \leq (k+1)(R(\underbrace{3, 3, \dots, 3}_k) - 1) + 2$$

Use this result to obtain an upper bound for

$$R(\underbrace{3, 3, \dots, 3}_n)$$

Define

$$f(k) = R(\underbrace{3, 3, 3 \dots, 3}_k)$$

Then we must prove $f(k+1) \leq (k+1)(f(k) - 1) + 2$.

$f(k)$ is defined as the smallest number of elements such that a group size 3 all with equal coloring is guaranteed to exist. Therefore, a graph with $f(k) - 1$ elements does not guarantee all colorings satisfy the group size 3 existing.

Let's now construct a new graph with $k+1$ colors $c_1, c_2, \dots, c_{k+1}$.

Consider creating a graph S of size $|S| = (k+1)(f(k) - 1)$. We can divide this graph into $k+1$ subgraphs $s_1, s_2, \dots, s_{k+1}$, all size $f(k) - 1$. For any subgraph s_i , color all its connections all colors except the color c_i . This means each subgraph is colored with k colors total. Therefore, each subgraph will not be guaranteed to create a group size 3 for which all connections are of the same color.

We now demonstrate that adding 2 elements to this graph guarantees at least one group of size 3 becomes guaranteed.

Consider first placing an additional element e_1 to graph S . To avoid creating a 3 group, we color all its connections to elements from subgraph s_i with color c_i . Since none of the connections inside subgraph s_i are colored c_i , we are guaranteed to not create any 3 groups.

Now consider placing another element e_2 to the graph $S + e_1$. To avoid creating a 3 group with any of the s_i , subgraphs, all connections from e_2 to the subgraph s_i are colored with c_i . After all such colorings there is one connection left, the one from e_1 to e_2 . If we color this connection with some color c_i , we create a 3 group with any 2 nodes belonging to the s_i subgraph:

$$A \in s_i : e_1 \xrightarrow{c_i} A, e_2 \xrightarrow{c_i} A, e_1 \xrightarrow{c_i} e_2$$

Thus, no matter how the graph of size $(k+1)(f(k) - 1) + 2$ gets colored, there is always a group of 3 whose connections are all the same color. $\square$

Now to obtain the upper bound for $R(\underbrace{3, 3, \dots, 3}_n) = f(k)$, the recursive formula gets expanded:

$$\begin{aligned}
f(1) &= 3 \\
f(k) &\leq k(f(k-1) - 1) + 2 \\
f(k) &\leq k((k-1)(f(k-2) - 1) + 2) + 2 \\
f(k) &\leq k((k-1)((k-2)(f(k-3) - 1) + 2) + 2) + 2 \\
&\rightarrow \\
f(k) &\leq \dots((k - (k-2))(f(k - (k-1)) - 1) + 2)\dots \\
f(k) &\leq \dots((2)(f(1) - 1) + 2)\dots \\
&\rightarrow \\
f(k) &\leq k((k-1)((k-2)((\dots((2)(f(1) - 1) + 2)\dots) + 2) + 2) + 2) + 2 \\
f(k) &\leq k((k-1)((k-2)((\dots((2)(2) + 2)\dots) + 2) + 2) + 2) + 2 \\
&\rightarrow \\
f(k) &\leq 2 * k(k-1)(k-2)(k-3)\dots(2) + \\
&\quad 2 * k(k-1)(k-2)(k-3)\dots(3) + \\
&\quad 2 * k(k-1)(k-2)(k-3)\dots(4) + \\
&\quad \dots + 2k + 2 \\
&\rightarrow \\
f(k) &\leq 2 \left(k! + \frac{k!}{2!} + \frac{k!}{3!} + \dots + \frac{k!}{(k-1)!} + 1 \right) \\
&\rightarrow \\
\boxed{f(k) \leq 2 \sum_{n=1}^k \frac{k!}{n!}}
\end{aligned}$$

23

The line segments joining 10 points are arbitrarily colored red or blue. Prove that there must exist three points such that the three line segments joining them are all red, or four points such that the six line segments joining them are all blue (that is, $R(3, 4) \leq 10$).

Consider some element A . It has 9 connections to all other elements, and all connections are either red or blue. We can consider the group of all elements with blue connections to A and the group of all elements with red connections to A .

Let n denote the number of elements that have red connections with A . All connections inside this group must be blue, since any red connection would complete a triangle with A . Therefore, $n \leq 3$, since any larger n will create a group of 4 with all blue connections.

Since $n \leq 3$, at least 6 elements must be connected to A as blue. Now the 6 remaining elements must be connected to avoid creating any red or blue triangles, since a red triangle satisfies the condition, and any blue triangle also creates a 4 group since all connections to A are blue. Since $R(6, 6) = 3$, any group of 3 people is guaranteed to make either a red or blue triangle, meaning in all graphs size 10 the ramsey condition is satisfied. $\square$

24

Let q_3 and t be positive integers with $q_3 \geq t$. Determine the Ramsey number $r_t(t, t, q_3)$.

Consider a graph of size $n \geq q_3$. We must color all t -subsets of the graph with colors c_1, c_2, c_3 , such that no group of t -vertices all colored c_1 or c_2 exist, and no group of q_3 -vertices all colored c_3 exist.

Coloring any t -subset with color c_1 immediately creates a group of t elements, all with color c_1 . In the case that any t -subset is colored c_2 , a group of t elements all colored c_2 is created as well. Therefore, no t -subset can be colored either c_1 or c_2 . However, this implies that every t -subset is colored c_3 , meaning all n elements all have their t -subsets colored c_3 . Since $n \geq q_3$, the condition is still satisfied. Therefore, $r_t(t, t, q_3) \geq q_3$.

In the case that $n < q_3$, the group created by coloring all t -subsets q_3 is still n elements in size. However, this size is smaller than q_3 , so the ramsey condition is not satisfied. Therefore, $r_t(t, t, q_3) \geq q_3$.

Thus, $r_t(t, t, q_3) = q_3$. $\square$

25

Let $q_1, q_2, \dots, q_k, t$ be positive integers, where $q_1 \geq t, q_2 \geq t, \dots, q_k \geq t$. Let m be the largest of $q_1, q_2, \dots, q_k$. Show that

$$r_t(m, m, \dots, m) \geq r_t(q_1, q_2, \dots, q_k)$$

Conclude that, to prove Ramsey's theorem, it is enough to prove it in the case that $q_1 = q_2 = \dots = q_k$.

Let $M = r_t(m, m, \dots, m)$. By definition, all graphs of size M that get colored with k colors become will have a group of at least m elements, whose t -subsets are all colored the same. Now instead consider $r_t(q_1, q_2, \dots, q_k)$, where $q_i \leq m$. Any graph of size M is guaranteed to satisfy the ramsey condition for a group of m size, however any group of m size also satisfies a group of smaller size, say for some $q_i \leq m$. Therefore, all graphs that satisfy $r_t(m, m, \dots, m)$ also satisfy $r_t(q_1, q_2, \dots, q_k)$. Since r_t is minimizing the largest it can equal is $r_t(m, m, \dots, m)$:

$$r_t(m, m, \dots, m) \geq r_t(q_1, q_2, \dots, q_k)$$

Ramsey's theorem states that some p exists such that

$$K_p^t \rightarrow K_{q_1}^t, K_{q_2}^t, \dots, K_{q_k}^t$$

Consider the case $q_1 = q_2 = \dots = q_k = m$. In this case, any valid p is also valid for the cases where $m \geq q_i$ for any i , since a valid group of m also satisfies a group of size smaller than m . Therefore, it suffices to prove the case $q_1 = q_2 = \dots = q_k = m$ to prove for any q_i .

Chapter 11 problems

A

Prove that there is no graph with an odd number of vertices of odd degree.

Each vertex must have at least 1 edge, since having 0 edges would make it of even degree. We can achieve this for any graph of size k by making $\frac{k}{2}$ pairs of connected vertices. This makes the degree of all vertices 1. This only works when k is even.

Consider the case when k is odd. $\lfloor \frac{k}{2} \rfloor$ pairs of vertices are made, and one vertex a is left with no connections. To make it degree 1 it gets connected to some vertex b . However, this makes b of degree 2, so to make it odd again b gets connected to another vertex. This process repeats until one node has degree 2 and every other one has degree 3, which is equivalent to the case of a with no connections and every other with 1. Continuing this pattern the graph ends up having every vertex connected to every other. In this case, each vertex has a degree of $k - 1$, still resulting in an even degree. $\square$

B

Let $P = (v_1, v_2, \dots, v_k)$ be a path with $k \geq 2$. Let $\tau : \{v_1, v_2, \dots, v_k\} \rightarrow \{1, 2\}$ be a mapping such that $\tau(v_1) = 1$ and $\tau(v_k) = 2$. Prove that P has an odd number of rainbow edges under τ , where an edge $\{u, v\}$ of P is rainbow under τ if $\tau(u) \neq \tau(v)$.

Consider $\tau(v_i) = 1, i < k$. In this case there is only 1 rainbow edge: the edge $\{v_{k-1}, v_k\}$. Now consider the case where $\tau(v_j) = 2, j < k - 1$ and $\tau(v_i) = 1, i < k, i \neq j$. In this case there are 3 rainbow edges: $\{v_{j-1}, v_j\}, \{v_j, v_{j+1}\}, \{v_{k-1}, v_k\}$.

In general, coloring some v_i either doesn't increase the number of rainbow edges, or it increases the rainbow edges by two.

Consider the case where a rainbow edge is $\{v_i, v_{i+1}\}$, and $\tau(v_{i-1}) = \tau(v_i)$. Let's say $\tau(v_i) = 1$ and $\tau(v_{i+1}) = 2$. Now if v_i gets changed to now be $\tau(v_i) = 2$, the # of rainbow edges doesn't change, only the edge itself does (changes from $\{v_i, v_{i+1}\}$ to $\{v_{i-1}, v_i\}$).

Now consider the opposite case where $\tau(i - 1) = \tau(i) = \tau(i + 1)$. Changing $\tau(v_i)$ introduces two new rainbow edges: $\{v_{i-1}, v_i\}$ and $\{v_i, v_{i+1}\}$.

In other words, changing the color of any v_i either doesn't change the # of rainbow edges, or it increase it by 2. $\tau(v_1) \neq \tau(v_k)$ implies there must be at least 1 rainbow edge. Thus, the # of rainbow edges is always odd. $\square$